HORIZON-GATED LATENT PREDICTION: DISENTANGLING PERSISTENT FACTORS IN TIME SERIES

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ABSTRACT

A long-lasting factor in a time series need not be a slow one: a factor can persist while riding on a nonlinear statistic such as an oscillation's frequency, where no low-pass reading reaches it. What persistent factors of either kind share is that they stay useful for predicting the distant future, while short-lived ones do not. We introduce Horizon-Gated Latent Prediction (HGLP), a self-supervised representation learning method for time series that disentangles long-lasting *persistent* factors from short-lived *transient* ones by gating the latent space as a function of the prediction horizon. A soft gate prevents long-horizon predictions from using a designated subspace (z_{mix}), forcing the model to route long-range predictive information through a specific block (z_{per}); a cross-covariance penalty and a per-block LayerNorm further encourage separation. We also propose a block-by-factor evaluation protocol and a single-number Separation Index (SEP), and demonstrate that the method separates factors on synthetic data and, more mixedly, on two real datasets (PTB-XL ECG, HAPT activities), with ablations showing the necessity of both the gate and a latent target. The penalty's one free coefficient can be set with no task labels at all.

1 INTRODUCTION

A sensor stream that runs unattended almost always carries two things at once. A wearable accelerometer records both what the wearer is doing and how the device happens to be shaking in this instant; a machine's vibration records both its accumulated wear and the load of the current second. The quantity anyone acts on is the first of each pair — the state that lasts — and the second changes far faster than any decision taken from it. This is the ordinary situation in condition monitoring, activity recognition, sleep staging and health-state estimation, and it is why the two are worth keeping in separate coordinates. A named block buys three things: interpretation gets a fixed address rather than a post-hoc story, the price of using only that block becomes a measurable number, and scarce labels can be spent on a narrow subspace instead of the whole embedding.

Why the separation has to happen during training. An embedding can always be rotated afterwards, and unsupervised unmixings — principal component analysis, slow feature analysis (SFA), independent component analysis — do exactly that at a fraction of the cost, so this is the sharpest objection to anything done at training time. Our answer is that a post-hoc method must rank directions by a quantity computable from a frozen embedding: variance, slowness, non-Gaussianity, or predictability under a linear autoregressive model (Richthofer & Wiskott, 2013). **None of those is persistence.** Each stands in for it, and the substitution stays invisible until the surrogate and persistence disagree — on our synthetic data, built so they do, a variance ranking fails outright (SEP 0.000) and slowness-based unmixing reaches barely half our score. The horizon is not a property of the embedding at all but of the predictive objective: free while training, gone once training ends. A rotation is also conservative, moving information between coordinates without changing how much of it the encoder represents, whereas training changes that — our complement block recovers the transient factor better than SFA's complement in 5 of six settings. The counterweight belongs in the same paragraph: where the surrogate and persistence do agree, post-hoc rotation is as good or better, and on both of our real datasets they do, which is why SFA leads us there.

Persistence is not confined to the slow band. Separating "slow" from "fast" can mean separating the *signal*, decomposing a waveform into trend and fluctuation (Cleveland et al., 1990; Huang et al., 1998), or separating the *factors* that generated it (Hyvärinen & Morioka, 2016) — a patient's diagnosis, or the fact that a person is walking. The two readings are routinely treated as one, and they diverge in exactly one direction. Fixing the term: a factor is *persistent* if it still contributes to prediction Δ steps ahead. The slow part of the signal is then necessarily persistent, since a component confined to low frequencies has slowly decaying autocorrelation (Khinchin, 1934) and remaining autocorrelation is a contribution to distant prediction. The converse fails, because low-pass filtering is linear in the signal while a factor may ride on a nonlinear statistic of it. Engineering practice knows this: in rolling-element bearing diagnostics the fault frequencies do not stand out in the spectrum of the raw vibration and become apparent only after demodulation, in the spectrum of the Hilbert envelope (Randall & Antoni, 2011); in frequency-modulated radio the message lives in the instantaneous frequency of a fast carrier, where no low-pass of the raw waveform reaches it. Our synthetic generator is built on that second shape deliberately, and Section 3.1 shows what it costs a slowness criterion. This is an existence claim, not a claim about how often such factors arise.

One criterion covers both kinds. So a method keyed to the slow band reaches persistent-and-slow factors and misses persistent-but-not-slow ones. What both kinds share is the property they were defined by: **both remain useful for predicting the distant future, and transient factors do not.** That criterion is controlled by a variable prediction-based learning already contains — the horizon Δ, since pushing it out leaves only long-persisting factors useful (Bialek et al., 2001; Creutzig & Sprekeler, 2008) — and nothing in that argument refers to which band the factor occupies. We gate that axis. HGLP places a single soft threshold on the horizon: beyond it the predictor may no longer read a designated block, so the only remaining route to lowering long-range loss is to write the current state into the block that is left. Because the latent space is rotation-symmetric, the persistent factor can be read from any 16 dimensions, and our claim is therefore not that it *gathers* in z_{per} but that the transient factor is *emptied out* of it; the complement z_{mix} is left unconstrained and keeps both (Section 3.2).

Contributions. (i) An inductive bias that gates an axis already present in any predictive learner, splitting the persistent factor into a designatable block, together with a criterion that sets the one coefficient governing that split *without task labels*. (ii) An evaluation protocol that scores the separation from both sides — a block $\times$ factor matrix — and a single summarizing index, SEP. (iii) Measurements on synthetic data, PTB-XL and HAPT, including an audit of where the method fails. The results cut both ways. On synthetic data we lead label-free post-hoc rotation on both loss families (SEP 0.849 against SFA's 0.388); on real data we trail SFA in three of four settings by 0.031–0.072. Splitting the index into its three terms locates that shortfall entirely on *exclusion*, the one term our objective cannot enforce, and replacing the linear probe with a small neural one — given equally to every baseline — collapses that same term and no other. The separation also costs -0.111 macro-F1 downstream on HAPT, which we quantify rather than omit.

2 Related Work

2.1 Classical Methods

Classical methods apply a *fixed transform* to the raw signal to read out its factors, and the dominant line has been the search for slowly varying components, since the slowly varying part of a signal reflects the factors that persist. The most direct approach splits the waveform itself: STL decomposes a series into seasonal and trend components (Cleveland et al., 1990), and EMD extracts oscillatory modes empirically (Huang et al., 1998). These deliver *components of the signal*, not the factors that generated it.

Another strand optimizes the slowness of the output directly. Slow Feature Analysis (SFA) minimizes the temporal derivative of the output under unit variance and pairwise decorrelation constraints, and, since the input is expanded into low-order polynomials, it also extracts nonlinear features (Wiskott & Sejnowski, 2002). Strengthening decorrelation into statistical independence makes the slow components separate from one another too, combining slowness with independent component analysis (Blaschke et al., 2007). We use these two variants as our label-free post-hoc rotation baselines. A further strand changes the criterion: Predictable Feature Analysis (PFA) keeps SFA's extraction scheme but selects the most predictable features instead of the slowest, predictability being how well the next value is approximated from the most recent p values (Richthofer & Wiskott, 2013). Yet another changes where the signal is read: the Fourier transform reaches factors riding on the *modulation* rather than the values, and in rotating-machinery diagnosis a bearing fault's characteristic frequencies appear not in the raw spectrum but in the *spectrum of the envelope* obtained through the Hilbert transform (Randall & Antoni, 2011).

What this family has in common is that *the transform is not learned*: which statistics and bands to keep are fixed in advance by a human, and the projection is refitted over the whole signal whenever the domain changes.

2.2 Self-Supervised Learning for Time Series

Time-series representation learning trains an encoder mapping a signal segment to a vector without labels, taking its learning signal from the temporal structure of the data—neighboring segments should be similar, or what is observed now should predict what comes shortly after. The family splits into three groups by the *form of the target*.

The *reconstruction family* is the largest and effectively the default. Autoencoders pass a signal through a narrow bottleneck and restore it, the remaining error itself being useful for anomaly detection, imputation, and forecasting pretraining. Variational autoencoders add a prior over the latent variables, and that prior becomes a *handle for inducing structure in the latent space*: a representative variant weights the KL term to push the coordinates toward independence (Higgins et al., 2017), a lineage leading to Section 2.3. Masked pretraining, which reconstructs randomly masked patches, has since become established (Nie et al., 2023).

The *contrastive family* decides which pairs count as similar and pulls and pushes the representations accordingly. CPC predicts the future contrastively in the latent space (van den Oord et al., 2018), and later time-series methods refined those pairs: contrasting hierarchically over augmented context views (Yue et al., 2022), or using local smoothness to define stationary temporal neighborhoods (Tonekaboni et al., 2021). Their weakness is that what gets discarded is hidden in the pair-selection rule.

The *latent-prediction family* takes the representation itself, not the raw signal, as the target (LeCun, 2022; Assran et al., 2023). Its time-series variants differ in where the target is read: from masked patches (Ennadir et al., 2025), from the next latent token (Chemeris et al., 2026), or from an attention-pooled summary of the whole future interval (Petersen et al., 2026).

This work belongs to the latent-prediction family but uses *both* a regression-form (L_1) and a contrastive-form (InfoNCE) loss, reporting the two stems together to show that the horizon gate does not rely on a particular loss form. All three groups *learn* the representation, but the objective does not determine what each coordinate means.

2.3 Disentangled Representations for Time Series

Work here has been most successful at separating *factors constant over an entire sequence* from the rest, because invariance transfers directly into an objective. FHVAE separates what stays fixed within a sequence from what changes across segments by assigning different priors (Hsu et al., 2017), and the disentangled sequential autoencoder splits the latent into static and dynamic parts (Li & Mandt, 2018). On identifiability, exploiting non-stationarity by learning to distinguish time segments makes a nonlinear ICA model identifiable *up to a pointwise transformation of each factor* when combined with linear ICA (Hyvärinen & Morioka, 2016), and an approach placing a *sparse* prior on the change between adjacent observations followed (Klindt et al., 2021). That models cannot be selected without labels (Locatello et al., 2019) applies equally to our hyperparameters, and we address it in the limitations.

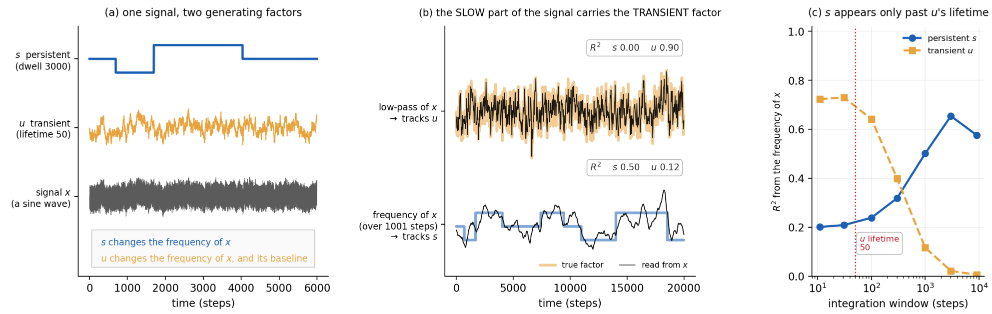

Figure 1: **Spectral slowness is not factor persistence.** One synthetic series (400,000 steps, a single seed): the persistent factor s switches every 3,000 steps among three frequencies 3% apart, and the transient factor u is an Ornstein–Uhlenbeck process of lifetime 50. Autocorrelation half-lives are 1,859 and 51 steps, so the two timescales are well separated. **(a)** The two factors and the signal they generate. s changes only the frequency of x; u changes its frequency and its baseline. **(b)** Reading the slow part of x recovers the transient factor rather than the persistent one (R^2 0.90 against 0.00), while reading the frequency of x over a 1,001-step window recovers the persistent one (R^2 0.50). **(c)** R^2 for both factors against the width of that window. The persistent factor appears only once the window is tens of times u's lifetime, and falls off again when the window grows wide enough to blur s itself.

What this family leaves empty is *factors that vary slowly but are not constant*: the static/non-static dichotomy pushes them into the "dynamic" side and separates them no further. None of the three strands creates coordinate-block structure of the kind we seek; this work fills that place with the axis of the prediction horizon.

3 Background

3.1 Where persistence is not slowness

If persistence and spectral slowness always agreed, a horizon gate would be an expensive way to reach what a low-pass filter already gives. To test the method where they disagree we need data in which the disagreement is a fact of construction rather than a hope, so we build one.

A three-state Markov persistent factor s (mean dwell 3,000 steps) sets the carrier frequency of an oscillation, spaced $\pm 3\%$ around 0.100; a transient Ornstein–Uhlenbeck factor u (autocorrelation time 50 steps) modulates that instantaneous frequency and adds to the signal. Amplitude is identical across regimes, so a regime change alters *only how fast the signal oscillates*. The two timescales are cleanly separated — autocorrelation half-lives 1,859 and 51 steps — so s is persistent by any reading. Yet it is absent from the slow part of the signal: a low-pass reading ($f < 0.02$) recovers the *transient* factor at $R^2 = 0.90$ and the persistent one at $R^2 < 10^{-5}$ (Fig. 1b). The persistent factor is carried by the fastest-changing part of the signal and the transient factor enters as the slower additive term, which is the exact reversal of what a slowness criterion assumes.

The persistent factor is still readable, but only through a nonlinear statistic and only over a wide enough window: reading the instantaneous frequency over 1,001 steps recovers it at $R^2 = 0.50$, and the recovery appears only once the window is tens of times u's lifetime, falling away again when the window grows wide enough to blur s itself (Fig. 1c). This is the general shape of the situation, not an artifact of our generator: while a factor persists, what stays fixed is not the signal but its statistics, and those surface only through some function of the signal.

Difficulty is set by the spacing between regime frequencies and calibrated by a training-free FFT reader, so that the task is neither trivially solved nor unsolvable before any representation is learned. At $\pm 3\%$ the reader reaches accuracy 0.660 (chance 0.333); at $\pm 5\%$ it already reaches 0.784 before any training, and at $\pm 1\%$ it falls to 0.459, near chance. We use $\pm 3\%$ throughout.

3.2 Why near-horizon prediction needs both factors

The gate is one-sided: z_{per} is available at every horizon, while z_{mix} is withheld once Δ exceeds the threshold. The asymmetry is the design, and it follows from what prediction at each range actually requires.

At long range the transient factor is already forgotten, so withholding the block that carries it should cost nothing — that is the premise the gate rests on, and it is what makes the remaining route, writing the current state into z_{per}, the only way to lower the loss. At short range the requirement is different and, importantly, not symmetric. Predicting a few steps ahead needs the fast detail of the present *and* the current persistent state, because the near future is a continuation of both. A model that withheld z_{per} at short horizons would not gain exclusion; it would remove the block from most of its own training signal, since short horizons are where gradients are dense. We test that variant rather than assert it, in Appendix A.

Two consequences are worth stating before any result. First, **only the *availability* of z_{mix} is constrained, never its content.** Nothing in the objective asks z_{mix} to be free of the persistent factor, and it is not: across the six settings it carries the persistent factor at 0.707–0.879 (macro-F1 or accuracy, against chance levels of 0.167 to 0.500). This is why we call the block z_{mix} and not "z_{fast}" — the latter name would promise a symmetric split the method never claims.

Second, the separation the method can claim is one-directional (Fig. 2). The latent space is rotation-symmetric, so the persistent factor can be read out of any 16 dimensions and its presence in z_{per} is no evidence of anything. The claim is the other one: that the *transient* factor is emptied out of z_{per}. Everything we measure is built around that asymmetry, and so is the honest weakness of the method — the objective can force a factor to be available in a block, but it has no term that forces one out.

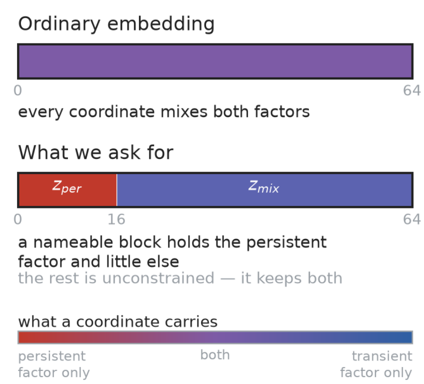

Figure 2: **What we ask of the embedding.** A nameable 16-dimensional block, z_{per}, emptied of the transient factor; the rest left unconstrained, keeping both. The claim is about what is *absent* from z_{per}, not about where the persistent factor can be found.

4 Method

4.1 Model

Encoder. One patch becomes one token (Fig. 3(a)) through a *single* linear projection; there is no codebook and no quantization. Channels are mixed at this first layer by design, since the factor of interest is sometimes defined by the relation between channels — HAPT posture is determined by how gravity distributes over the three axes. Tokens then pass through a stack of **causal transformer layers** with rotary position encoding, and a final linear layer produces the embedding, whose leading coordinates form z_{per} and the remainder z_{mix}. The last operation is a **per-block LayerNorm**, applied to the two blocks separately rather than to the embedding as a whole, for the reason given in Section 4.3. The **predictor** is a small MLP that sees only the embedding at one time point together with the horizon, which enters as a learned expansion of $\log_2 \Delta$; one predictor is conditioned on the horizon rather than one head per horizon, consistent with treating the horizon as a continuous axis.

No modification is made to the attention operation: the gate sits between encoder output and predictor input, the penalty in the loss, the per-block LayerNorm in the encoder's last line. This is a change of training procedure, not of architecture, so it attaches to any causal sequence encoder.

Target.

$$z^* = \text{sg}\big[\, f_{\bar\theta}\big(x_{[t+\Delta-w,\, t+\Delta]}\big)_{-1}\big], \qquad \bar\theta \leftarrow m\bar\theta + (1-m)\theta \tag{1}$$

The prediction target is not the raw signal but the representation an EMA copy of the encoder reads from a short window inside the future interval, with the gradient stopped. Bounding that window keeps the target's receptive field inside $(t,\, t+\Delta]$, so it never re-contains the anchor's own past

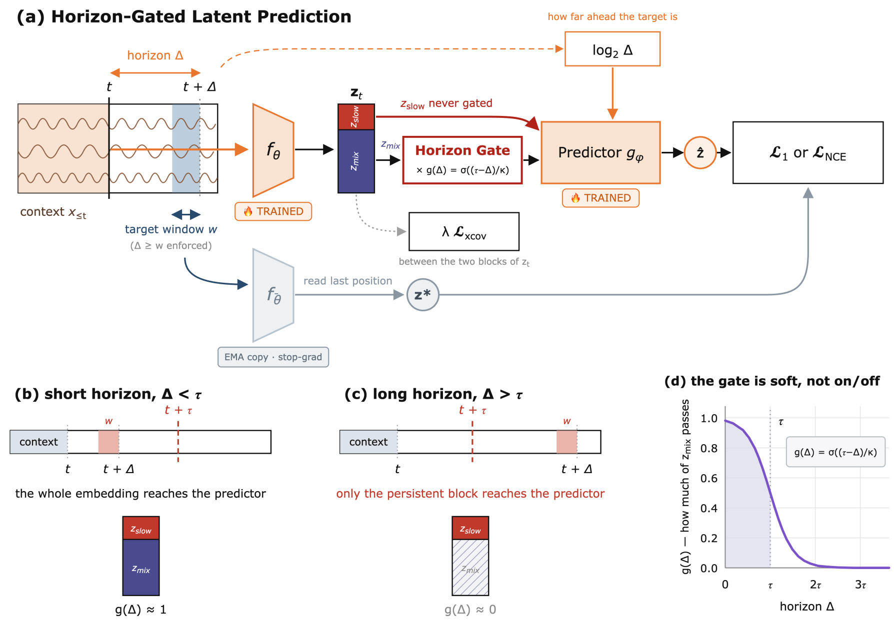

Figure 3: Horizon-gated latent prediction. **(a)** A causal encoder maps a window to an embedding z_t, split into a persistent block z_{per} (16 of 64 dimensions) and the rest z_{mix}. A predictor conditioned on the horizon Δ must match a target read at one position inside $(t,\, t+\Delta]$, produced by an EMA copy of the encoder. A cross-covariance penalty is applied to z_t before the gate. **(b, c)** The same pipeline at a short and a long horizon. z_{per} is never gated; only z_{mix} is, and only past τ, so at long horizons the only route to lowering the loss runs through z_{per}. **(d)** The gate is a sigmoid of width κ, not a switch. τ is set from the transient factor's lifetime and Δ_{max} is set separately, so the axis is drawn in units of τ rather than absolute horizons.

and the shortcut of matching the target from what is already known at the anchor is closed by construction. **The target never passes the gate and is always the full embedding**: at far horizons the model is asked to match the whole future embedding from the persistent block alone, and this asymmetry is the core of the mechanism.

Loss.

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda\,\mathcal{L}_{\text{xcov}}, \qquad \mathcal{L}_{\text{pred}} \in \{\mathcal{L}_1,\ \mathcal{L}_{\text{NCE}}\} \tag{2}$$

The two forms share backbone, gate and penalty and differ only in the loss layer. **HGLP-Reg** uses the L_1 distance; L_2's squared-error gradient is dominated by large residuals and buries a weak persistent signal, which made results swing across seeds. **HGLP-NCE** uses InfoNCE over cosine similarity with temperature 0.1, with one departure from the standard form: **samples from the same window are excluded from the negative pairs**, since they share the persistent factor and pushing them apart would teach the model to discard it.

4.2 The Horizon Gate

$$g(\Delta) = \sigma\!\left(\frac{\tau - \Delta}{\kappa}\right), \qquad \tilde{z}_t(\Delta) = \left[\, z_t^{\text{per}}\,;\; g(\Delta)\, z_t^{\text{mix}} \,\right] \tag{3}$$

Immediately before the predictor, the rear block is multiplied by $g(\Delta) \in (0, 1)$ (Fig. 3(b)–(d)): for $\Delta \ll \tau$ the embedding passes unchanged, for $\Delta \gg \tau$ the rear block is suppressed. *Dimensions are multiplied, not truncated*, so the predictor input is always D-dimensional; z_{per} *is never blocked*; and the gate is *a smooth curve, not a step*. That z_{per} is never blocked is a choice and not an omission: the symmetric alternative, switching z_{per} off below τ so that no horizon can deposit transient information in it, is the obvious way to buy the exclusion that this objective cannot enforce, and we test it in Appendix A. A step gate would flip horizons near τ between fully passed and fully blocked under the smallest movement of the threshold, whereas width κ gives them intermediate values. Since τ cannot be chosen accurately without labels (Section 7), this buffer is necessary rather than convenient, and in our runs a step gate lowered the separation metric on both loss forms (Appendix A).

Gradient split.

$$\frac{\partial \mathcal{L}}{\partial z_t^{\text{per}}} = \frac{\partial \mathcal{L}}{\partial \tilde{z}_t^{\text{per}}}, \qquad \frac{\partial \mathcal{L}}{\partial z_t^{\text{mix}}} = g(\Delta)\,\frac{\partial \mathcal{L}}{\partial \tilde{z}_t^{\text{mix}}} \tag{4}$$

Because the encoder is shared across horizons, this one line splits the training signal the two blocks receive: z_{per} is updated at every horizon, whereas z_{mix} is updated in proportion to $g(\Delta)$ and so effectively only at near horizons. Two consequences follow, and only one of them is guaranteed. **Availability is enforced**: a predictor cannot foresee when the persistent factor will next switch, so at long horizons it can lower the loss only by recording the current state, and the gate leaves z_{per} as the sole route for doing so; so whatever still contributes to prediction that far ahead must be readable there, and no predictor however large can restore information the gate has suppressed to zero. **Exclusion is not enforced**: z_{per} is still used at near horizons, where transient information lowers the loss, and no term of the objective pushes that information *out of* z_{per} — exclusion has to come from the penalty of Section 4.3. We therefore measure exclusion rather than assume it, and Section 5.4 follows that measurement through — it is where our shortfall against the strongest baseline turns out to sit, and how far it holds when the probe is no longer linear (Appendix B); and the claim is that the transient factor is emptied out of z_{per}, not that the persistent factor gathers there: with all mechanisms absent, the first 16 dimensions and a random 16 agree to within 0.015.

4.3 Pulling the Two Blocks Apart

Cross-covariance penalty.

$$\begin{aligned} C &= \tfrac{1}{N-1}\,\bar{Z}^{\top}_{(1:d_{\text{per}})}\,\bar{Z}_{(d_{\text{per}}+1:D)}, \\ \mathcal{L}_{\text{xcov}} &= \tfrac{1}{d_{\text{per}}(D-d_{\text{per}})}\textstyle\sum_{a,b} C_{ab}^2 \end{aligned} \tag{5}$$

The penalty pushes the two blocks to be linearly unpredictable from each other and **acts on the embedding before the gate**. It travels with the **per-block LayerNorm**: a single LayerNorm over the whole embedding gives the two blocks one shared mean and variance, recreating exactly the coupling the penalty is trying to remove. **"Mechanism-free" therefore means all three devices absent** — gate, penalty and per-block LayerNorm.

4.4 What We Measure

Reading the persistent factor from z_{per} is not by itself evidence of separation, since rotational symmetry means it is read about as well from any 16 dimensions. We therefore report a **block $\times$ factor matrix** scoring both blocks on both factors, plus a single-number summary **SEP**, which asks three questions.

1. **Inclusion — is the persistent factor in z_{per}?** We score how well the persistent factor is predicted from z_{per} alone.
2. **Allocation — is the transient factor in z_{mix}?** We score how well the transient proxy is recovered from z_{mix} alone.
3. **Exclusion — is the transient factor absent from z_{per}?** We measure how well the transient proxy can be recovered from z_{per}, and subtract that value from 1.

$$\text{SEP} = \underbrace{\big[s_{\text{per}}(z^{\text{per}})\big]_0^1}_{\text{inclusion}} \cdot \underbrace{\big[s_{\text{tra}}(z^{\text{mix}})\big]_0^1}_{\text{allocation}} \cdot \underbrace{\big[1 - s_{\text{tra}}(z^{\text{per}})\big]_0^1}_{\text{exclusion}} \tag{6}$$

$s_{\text{per}}(\cdot)$ scores reading the *persistent factor* from a subspace and $s_{\text{tra}}(\cdot)$ the *transient proxy*; $[\cdot]_0^1$ clips into $[0, 1]$. Classification scores already lie in this range, but R^2 goes negative when a prediction is

Table 1: The three datasets. Persistent factor = the factor we try to separate; transient proxy = a fast-varying quantity computed directly from the signal, not from labels. Chance level is $1/k$ for balanced macro-F1.

	Synthetic	PTB-XL	HAPT
Samples	10^6 steps	5,000 records (4,812 patients)	2,104 windows (30 subjects)
Sampling rate	—	100 Hz	50 Hz
Channels	1	1 (lead II)	3 (accel. x, y, z)
Patch	8 steps	10 samples (0.1 s)	4 samples (0.08 s)
Window length L	256 patches (2,048 steps)	100 patches (10 s)	256 patches (20.5 s)
Persistent factor	oscillation regime s (3 classes)	ECG diagnosis (normal/ abnormal, **2 classes**, constant per record)	physical activity (6 classes)
Transient proxy	OU factor u	instantaneous voltage	accel. magnitude $\|\|a\|\|$
Chance level	0.333	0.500	0.167

worse than the mean, and the clipping acts only there, preventing a negative value from flipping the sign of the product. There is no normalization by any denominator anywhere.

Exclusion is the term that carries the claim. Unlike the other two it measures *what is absent*, and the objective enforces inclusion but not exclusion, so it is precisely the quantity we have to measure rather than assume.

Probe protocol. With the encoder frozen we fit only a linear classifier or regressor. Splits are **per group** (subject for HAPT, patient for PTB-XL, a front/back time split for the synthetic data), evaluation windows are **non-overlapping** to prevent leakage (asserted in the code), and only positions after the **minimum context length (64 patches)** are scored, since with less context the encoder lacks the evidence to read the persistent state. A **random subspace** of the same width is scored as a control.

5 Experiments

Every setting below is one of six: three datasets crossed with the two loss stems, at $n = 5$ seeds unless stated. We ask four questions in order — what produces the separation, whether its one free coefficient can be set without labels, how it compares against label-free post-hoc rotation, and what it costs. Design choices that could have been made differently are tested in Appendix A.

5.1 Data and factors

We use one synthetic dataset of our own design (Section 3.1) and two public real-world sets, PTB-XL (Wagner et al., 2020) and HAPT (Reyes-Ortiz et al., 2016; 2015) (Table 1). In each, the *persistent factor* is the quantity we try to separate and the *transient proxy* is a fast-varying quantity computed directly from the signal, not from labels.

The synthetic generator and its calibration are described in Section 3.1; it is the one dataset in which persistence and slowness are made to disagree by construction. On the two real sets the table gives the sizes and the factors, and three choices are worth stating. **Splits are per group** — patient for PTB-XL, subject for HAPT — and normalization uses training-split statistics only. **Nothing is selected by label before training**: HAPT's transition and unlabeled segments, 24.7% of the total, are excluded from scoring but left in the pre-training input, since filtering them out would undermine the claim that training uses no labels. And the two sets sit at opposite ends of one axis: the PTB-XL diagnosis is constant over a whole record, whereas HAPT's activity changes within one, so windows are cut without consulting activity boundaries, labels are read per position, and 57% of windows cross a transition. PTB-XL also strains the rule our threshold comes from — the heartbeat is a regularly repeating component, so its autocorrelation oscillates instead of decaying and no lifetime can be read from it, and τ has to be set by hand there (Section 7).

Table 2: Separation index by mechanism. Cells are SEP (higher is better), $n = 5$ seeds, $\lambda = 4$. Bold marks the best cell in each row. The gate and the penalty are complementary: neither alone reaches what the two reach together.

Dataset	Stem	Neither	Gate only	xcov only	**Gate + xcov**
Synthetic	Reg	0.135	0.342	0.279	**0.408**
Synthetic	NCE	0.121	0.124	0.488	**0.749**
PTB-XL	Reg	0.375	0.404	0.312	**0.495**
PTB-XL	NCE	0.392	0.503	0.283	**0.567**
HAPT	Reg	0.521	0.537	0.236	**0.632**
HAPT	NCE	0.297	0.366	0.162	**0.515**

5.2 What creates the separation

We switch each of the two devices we add — the horizon gate and the cross-covariance penalty — on and off. The mechanism-free cell (Section 4.3) is the baseline in which only the coordinate split remains.

In Table 2 the model with both devices ranks first in all six settings, and neither device alone reaches that value. On the contrastive stem of the synthetic data the gate alone is indistinguishable from the mechanism-free case (0.124 against 0.121), yet adding that same gate to the penalty raises SEP from 0.488 to 0.749: the two devices need each other. The penalty alone is actively harmful on the real data, breaking down the inclusion term and falling below even the mechanism-free case.

The "Neither" cells take the first 16 of the 64 dimensions, and their SEP agrees in all six settings to within 0.015 with the SEP of 16 dimensions drawn at random from the same embedding. Without our devices, any part of the embedding holds the two factors to a similar degree, so coordinate position carries no meaning by itself; our contribution is to give those coordinates structure. Fig. 4 shows the same contrast qualitatively: with the mechanism the transient factor is unstructured in z_{per}, and without it that block is no different from any other.

Three further design choices could each have gone the other way, and none of them is left to argument. Replacing the smooth gate with a step at τ, gating z_{per} symmetrically as well, and predicting the raw signal instead of a latent target are each tested in Appendix A; all three are worse, and the latent target by a factor of two.

5.3 Setting the penalty weight without labels

λ was chosen by looking at downstream performance, which makes "our best λ separates well" partly circular. We therefore define a criterion that uses no task label at all, the **purity–persistence score**: z_{per} should be *pure*, holding none of the transient factor, and *persistent*, still predicting its own value far ahead.

$$\mathrm{PPS} = \underbrace{\left[1 - s_{\text{tra}}(z^{\text{per}})\right]_0^1}_{\text{purity}} \cdot \underbrace{s_{\text{self}}(z^{\text{per}};\, \Delta > \tau)}_{\text{persistence}}, \tag{7}$$

Purity is one minus the transient proxy's score from z_{per} — the exclusion term of Eq. 6, and the proxy is computed from the signal. Persistence is how well z_{per} predicts *its own* value more than τ ahead; neither the labels nor the target encoder enters. The product is required because each half alone has a degenerate maximum: an empty block is perfectly pure, a constant block is perfectly self-predicting, and multiplying kills both.

Selecting λ by PPS reproduces the label-selected value in all six settings (Table 3). On synthetic data the criterion is also doing real work rather than agreeing by luck: purity alone would have chosen $\lambda = 16$, close to the worst choice by the oracle (SEP 0.332 against 0.814 at $\lambda = 1$), and it is the persistence half, collapsing from 0.670 to 0.051 once $\lambda \geq 4$, that rules it out. A collapse monitor would not have caught this: effective rank *rises* from 4.3 to 12.3 over the same range, so the penalty is not emptying z_{per} but scattering it along the time axis.

The agreement should not be over-read. The margin between the top two PPS values is 0.111–0.234 on synthetic but only 0.004–0.023 on the real sets, and we did not store its seed-level spread, so the four real-data agreements are not distinguishable from chance at this sample size. The honest

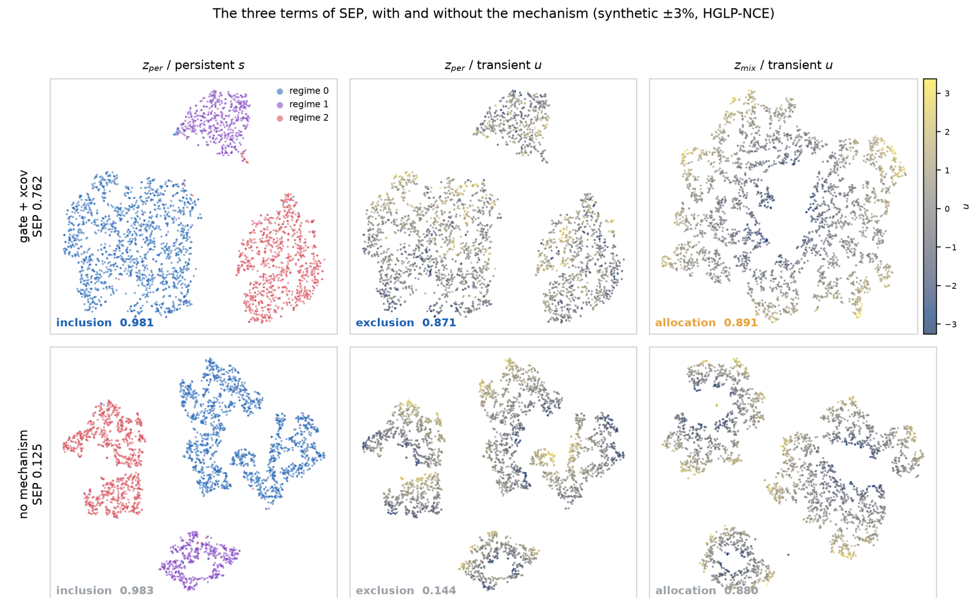

Figure 4: **The three terms of SEP, with and without the mechanism.** A t-SNE of one block per panel (synthetic $\pm 3\%$, HGLP-NCE, $\lambda = 4$, one seed), coloured by one factor; the columns are exactly the three terms of Eq. 6 and the bottom row is a model trained with no gate, no penalty and no per-block LayerNorm. Each panel carries its term's value, and **the whole gap between the rows is one column**: inclusion is 0.981 against 0.983 and allocation 0.891 against 0.880, both unchanged, while exclusion is 0.871 against 0.144 — SEP 0.762 against 0.125. The left column is therefore not a result: the regimes separate in z_{per} either way, as the random-subspace control above already said. What our devices change is the middle column. Read it by the colour scale rather than by eye, since the difference between a partly ordered and a fully scrambled scatter is exactly what a linear probe sees and a reader does not. The fourth combination, z_{mix} coloured by the persistent factor, is not a term of SEP and is unconstrained by design (Section 3.2), so it is omitted.

Table 3: Choosing λ without labels. $\text{PPS} = \text{purity} \times \text{persistence}$: the first factor is read from the transient proxy, which is computed from the signal, and the second asks how well z_{per} predicts *its own* value more than τ ahead. No task label enters either. The product is needed because each factor alone has a degenerate maximum — an empty block scores perfect exclusion, a constant block perfect self-prediction. The oracle column is the λ that maximizes SEP, which does use labels. $n = 3$ seeds; the margin is the gap between the top two PPS values, and small margins should not be read as confident agreement.

Dataset	Stem	λ by PPS (no labels)	λ by SEP (oracle)	Agreement	PPS margin
Synthetic	HGLP-Reg	1	1	match	0.234
Synthetic	HGLP-NCE	16	16	match	0.111
PTB-XL	HGLP-Reg	4	4	match	0.023
PTB-XL	HGLP-NCE	4	4	match	0.004
HAPT	HGLP-Reg	1	1	match	0.016
HAPT	HGLP-NCE	64	64	match	0.009

reading is that the criterion is verified on synthetic data, consistent on real data, and cheap to run wherever the transient proxy exists.

5.4 Against label-free post-hoc rotation

If an embedding trained without the gate is rotated post-hoc without labels — by principal component analysis, by slowness-ordered independent component analysis, or by slow feature analysis

Table 4: Our method against label-free post-hoc unmixings. Cells are SEP, $n = 5$ seeds. Our method has one knob, so we report its best point on the swept λ curve; each unmixing is a single fixed value. Bold marks the winner of each row. The verdict is three to three overall: two of two on synthetic, one of four on real data.

Dataset	Stem	Ours (best λ)	SFA	Verdict
Synthetic	Reg	**0.814** ($\lambda = 1$)	0.446	win
Synthetic	NCE	**0.849** ($\lambda = 16$)	0.388	win
PTB-XL	Reg	0.495	**0.567**	loss
PTB-XL	NCE	0.567	**0.599**	loss
HAPT	Reg	0.677	**0.717**	loss
HAPT	NCE	**0.626** ($\lambda = 64$)	0.527	win

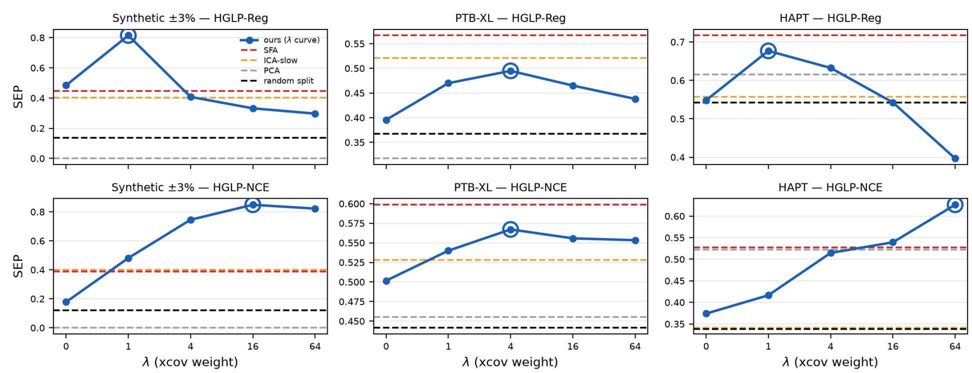

Figure 5: SEP against the penalty weight λ. $n = 5$ seeds. Our method traces a curve over $\lambda \in \{0, 1, 4, 16, 64\}$ with its best point ringed; each post-hoc unmixing has no knob and is a horizontal line. A win is our curve rising above the SFA line. Unlike an inclusion–exclusion plane, this view combines all three SEP terms, so it also shows the one real-data win, which the inclusion–exclusion plane cannot display.

(SFA) — does it reach the same separation? These methods need no horizon gate at all, so they are the sharpest objection to our method.

We have one knob and the post-hoc rotations have none, so we report the whole curve rather than a single point (Table 4, Fig. 5); hiding that asymmetry would not make the comparison fair. On the synthetic data the curve passes above every post-hoc rotation point, and principal component analysis fails completely (SEP 0.000) — direct evidence that a ranking by variance is not a ranking by persistence, since the transient factor crowds into the leading components and nothing is left in the complement. On the real data we do not lead. The verdict over all six settings is three wins to three losses, but split apart it is two of two on synthetic and three losses of four on real data, with losing margins of 0.031–0.072 against a winning margin of 0.099.

Where the shortfall sits. SEP is a product of three factors, so a single number cannot say which one is losing. Splitting it and pairing by seed gives an unusually clean answer (Appendix B, Table 10): inclusion is a draw in all six settings (-0.041 to $+0.102$), **every real-data loss is a loss on exclusion alone**, and every win is a win on allocation. The method does not degrade diffusely; it degrades on the one term the objective was never able to enforce (Section 4.2).

The exclusion we do achieve is a linear-level one. Every probe so far has been linear, and an unreadable direction is not the same as an absent one. Holding the features, the splits and the blocks fixed and replacing only the reading head with a one-hidden-layer MLP — given equally to SFA, PCA and ICA, since a harder test applied to us alone would not be a comparison — exactly one term collapses, and it is exclusion: averaged over four methods and four real settings, inclusion moves $+0.015$ and allocation $+0.087$, while exclusion falls -0.149 and as much as -0.530 in the worst row. The transient information is still present in z_{per}; the linear probe could not reach it. Our claim therefore narrows to a linear-level one — and the diagnosis above is not an artifact of weak probing,

Table 5: Degradation under domain shift: a difference of differences. Each cell is the drop in macro-F1 from the clean domain to the most perturbed one, for the gated block minus the same drop for a width-matched block of the mechanism-free model; paired by seed, $n = 5$. Negative means the gated block degrades less. **These are differences of drops, not differences in absolute performance** — the gated block does not overtake the mechanism-free one at any perturbation strength.

Setting	25 labels	100 labels	All labels
HAPT / Reg	$\mathbf{-0.113{\pm}0.023}$	$\mathbf{-0.096{\pm}0.029}$	$-0.053{\pm}0.056$
HAPT / NCE	$-0.017{\pm}0.072$	$+0.015{\pm}0.012$	$+0.007{\pm}0.016$
PTB-XL / Reg	$-0.044{\pm}0.048$	$-0.033{\pm}0.048$	$-0.024{\pm}0.035$
PTB-XL / NCE	$-0.036{\pm}0.032$	$-0.010{\pm}0.024$	$-0.009{\pm}0.024$
Synthetic / Reg	$-0.012{\pm}0.102$	$+0.043{\pm}0.099$	$+0.064{\pm}0.077$
Synthetic / NCE	$-0.044{\pm}0.069$	$-0.055{\pm}0.054$	$-0.045{\pm}0.055$

since SFA's lead on exclusion survives the stronger head and in places widens, while PCA and ICA degrade sharply. Appendix B gives the per-setting values and the caveats.

Why SFA leads where it leads. The explanation is that persistence and slowness mostly coincide on real data. On PTB-XL this is plain: the diagnosis is constant within a record, so its time derivative is zero, which is the limiting solution of the quantity SFA maximizes. On HAPT it is not constant within a window and we did not measure why the coincidence holds there, so that half of the explanation remains untested. What the synthetic data shows is the other side of the same statement: built so that the persistent factor is neither constant within a window nor present in the slow band, it is the one dataset on which the post-hoc route fails. What remains ours is that the coordinates are fixed during training, so no second fitting stage is needed at deployment — an operational difference, not a difference in the quality of the separation.

5.5 The cost of separation, and stability under domain shift

We measure two things: the downstream cost that accompanies the separation, and how far performance falls when a probe fitted with few labels is carried to a domain in which the transient factor has been perturbed. What we measure in the second case is not the difference in absolute performance between the two blocks but the difference in the drop each block suffers.

We measure the cost against the full embedding of a model trained with no mechanism at all rather than against our own z_{full}, since the latter mixes in the effect of block width. Read that way, the cost is effectively zero on synthetic data and PTB-XL, and only HAPT shows a substantial value (-0.111 on the regression stem, -0.072 on the contrastive stem). HAPT is the only dataset in which the persistent factor is not constant within the window: 57% of its windows straddle an activity transition.

The perturbed domain is made by multiplying the evaluation windows by a gain ($x \mapsto x \cdot (1 + s)$, maximum perturbation $s = 2$); this is a bijection and does not destroy the labels of the persistent factor — even at maximum perturbation the HAPT activity labels survive at macro-F1 0.696 (chance 0.167). The encoder was trained on the clean domain alone, and the probe is fitted only on clean data with few labels (25 or 100); only the evaluation runs separately on clean and perturbed windows, so the probe never trains on the perturbation it is scored on.

On that same HAPT the gated block has the smallest drop under shift: at 25 labels its drop is 0.113 smaller than that of the width-matched mechanism-free block (Table 5), five times the seed standard deviation. This is only a difference of drops. At no perturbation strength we tested does absolute performance overtake that model, and the effect exceeds the error bars in only one of the six settings, the HAPT regression stem; in the remaining five it lies within the noise, though by sign alone 17 of the 24 cells are negative.

The two results interlock: HAPT is both the dataset with the largest downstream cost and the only one with clear stability under domain shift, two sides of one phenomenon. The gate trades absolute performance for stability under domain shift; the two cannot be had together. A use that does not go through a classifier at all — reading the trajectory of z_{per} over time as a state indicator — is shown in Appendix C, and is the direction in which we think the separated block earns its cost.

6 CONCLUSIONS

The horizon gate and the cross-covariance penalty split the persistent factor into a designated coordinate block, and across all six settings formed by three datasets and two loss families they jointly produce a separation that neither device reaches alone. The separation requires a latent target: switching only the prediction target to the raw signal halves the separation score, and what breaks down is precisely the inclusion of the persistent factor. On synthetic data we exceed label-free post-hoc rotation for both loss families, but on real data we fall short of Slow Feature Analysis, losing three of the four real settings, and the separation costs 0.111 macro-F1 downstream on HAPT. The gate trades absolute performance for stability under domain shift.

Three measurements sharpen that comparison. Splitting SEP into its three factors shows that the shortfall against SFA is confined to the exclusion term — the one term the objective cannot enforce — while inclusion is a draw everywhere; that shortfall is not an artifact of linear probing, since it survives and in places widens under a non-linear reader which also collapses our own exclusion; and the coefficient that governs exclusion can be selected without task labels, agreeing with the label-based choice in all six settings.

6.1 LIMITATIONS

The exclusion we demonstrate holds at the linear level only. The objective enforces availability but not exclusion, and Section 5.4 shows that this is not a formality: swapping the linear probe for a small MLP, with the same head given to every baseline, leaves inclusion and allocation intact and collapses exclusion alone, by -0.149 on average and -0.530 at worst. The transient factor is not absent from z_{per}; it is linearly unreadable there. A structural repair does not help either — a gate that also switches z_{per} off below τ is worse in 4 of six settings. What we can claim is that the method fails where its own analysis said it would, and nowhere else.

Most hyperparameters still cannot be chosen without labels. The block width, the target width, the patch size and the horizon range have no derivation rule and were set by downstream performance, a known difficulty in this literature (Locatello et al., 2019); the target width in particular governs how much of the transient factor is averaged away inside the target, and we never swept it. The cross-covariance coefficient is the exception, and a partial one: the label-free purity–persistence score (Eq. 7) agrees with the label-based choice in all six settings, but decisively only on synthetic data, where the margin to the runner-up is 0.111–0.234 against 0.004–0.023 on real data, and we did not store the seed-level spread that would settle the four real agreements. On a fourth dataset not reported here the criterion selected a different value from the label-based one; that analysis is unfinished — one configuration axis was never tested — and we report neither its results nor its disagreement as evidence.

We did not demonstrate a practical benefit. The perturbed-domain effect clears the error bars in only one of six settings, and even there absolute performance never overtakes the mechanism-free embedding: the gated block degrades less, which is not the same as being better.

Our axis selects persistence, not relevance, and the case for it rests on data we built. A persistent nuisance factor — the low-frequency drift respiration and electrode motion produce in an electrocardiogram — enters z_{per} by the same criterion, and which persistent factors matter remains a human judgment. The claim that a factor can be persistent yet absent from the slow part of the signal likewise rests on a single synthetic dataset: one counterexample does refute a universal proposition, but we designed this one ourselves, and neither PTB-XL nor HAPT exhibits the case.

6.2 FUTURE WORK

We compared only against label-free post-hoc rotation, so methods that place factor separation explicitly in the objective remain to be compared — as does predictable feature analysis (Richthofer & Wiskott, 2013), the post-hoc criterion closest to persistence and the one our argument in Section 5.4 most owes a measurement. Validation is also needed on data whose persistent factor varies within a recording while staying constant within a window: PTB-XL fails the first property and HAPT the second, whereas sleep-stage classification has both. We have that fourth dataset in hand and the analysis is under way; it is the one referred to above, and it is unreported because its window-length axis has not been tested. Finally, reading the trajectory of the separated block as a state indicator rather than as classifier input (Appendix C) would address the third limitation directly.

7 Reproducibility Statement

What is fixed and what is swept. Every result is reported over five seeds unless the caption says otherwise; the seed varies both the data draw and the initialization, so the error bars are not initialization-only. Splits are **per group** — subject for HAPT, patient for PTB-XL, a front/back time split for the synthetic data — evaluation windows are non-overlapping, and the code asserts both. Only positions after the minimum context length of 64 patches are scored. Encoders are frozen for every evaluation in this paper; nothing downstream is fine-tuned.

From stored runs to the numbers printed here. Each experiment writes a JSON file, and every table and every figure number in this paper is regenerated from those files by a script rather than transcribed. The tables of Sections 5.3 and 5.4 and of Appendix A come from one generator, which also emits the figures quoted in running text as macros, so a table and the sentence beside it cannot drift apart. The two datasets are public: PTB-XL (Wagner et al., 2020) and HAPT (Reyes-Ortiz et al., 2016; 2015). The synthetic generator is specified completely by Section 3.1 — three-state Markov factor, mean dwell 3,000 steps, carrier 0.100 spaced $\pm 3\%$, Ornstein–Uhlenbeck transient of autocorrelation time 50 — and needs no download.

Hand-picked values, named as such. Most knobs of this method cannot be chosen without labels. **The block width, the target window, the patch size and the range of training horizons have no derivation**, and we set them by looking at downstream performance. The cross-covariance weight λ was chosen the same way, and Section 5.3 then shows that a label-free criterion recovers the same value; we report both, and the label-free route is the one we would use on new data.

The threshold τ is a partial exception in the other direction. It can be initialized from the transient factor's measured autocorrelation time, but the rule fails on signals with a regularly repeating component: on PTB-XL the per-record estimates spread over a factor of 499 and the rule's output falls outside the training horizon range, so the value we use there is an arbitrary choice rather than the rule's output. The one necessary condition we can state is that the trained horizons must straddle the threshold, $\Delta_{\min} < \tau < \Delta_{\max}$, or the gate never opens or never closes.

One quantity does not depend on labels at all: the exclusion term is measured from the transient proxy, which is computed directly from the signal. A failed separation is therefore detectable without task labels, whatever the knobs were set to.

8 What in this report is new to GENE 501

An earlier version of this work was written for another course. That version proposed the horizon gate, the cross-covariance penalty and the SEP protocol, and reported the mechanism ablation, the post-hoc rotation comparison, and the downstream cost and shift results. Three things in this report are new, and they are of one kind: each takes a sentence the earlier version had to assert and turns it into a measurement.

A method change: the penalty weight is now chosen without labels. The earlier version set λ by downstream performance and listed that among its limitations. Section 5.3 defines the purity–persistence score, built from the transient proxy and long-horizon self-prediction, neither of which uses a task label, and shows it recovers the label-selected value in all six settings. This is the only change to the method itself.

Design choices, ablated instead of argued. The earlier version justified three choices in prose. Appendix A tests them: the step gate against the smooth one and the symmetric gate against the one-sided one are both new here, and the target's receptive-field bound is measured for the first time

— that pilot predates this report but was never reported, and the earlier version stated the bound as holding "by construction" with nothing behind it.

The method's limit, quantified. The earlier version conceded in words that its objective cannot enforce exclusion and that its probes are linear. Section 5.4 and Appendix B turn both concessions into numbers: decomposing SEP locates every real-data loss on the exclusion term alone, and replacing the linear probe with a small MLP — given equally to every baseline — collapses that term and no other. The result is not favourable to the method, which is the point: it says the failure is where the theory predicted rather than anywhere else, and it fixes how far the separation claim may be read.

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Table 6: A step gate against the smooth one. The smooth gate is $g(\Delta) = \sigma((\tau - \Delta)/\kappa)$; the step version replaces it with $\mathbf{1}[\Delta < \tau]$, which zeroes z_{mix}'s gradient outright beyond τ instead of passing a partial one. Cells are SEP on synthetic, $n = 5$ seeds, all else held fixed. The step form is worse on both stems, which is why τ's width κ is part of the method rather than a smoothing convenience.

Stem	Smooth gate (ours)	Step gate	Difference
HGLP-Reg	**0.408**	0.353	-0.056
HGLP-NCE	**0.749**	0.684	-0.066

Table 7: Can exclusion be bought structurally? The symmetric variant multiplies z_{per} by $1 - g(\Delta)$, switching it off below τ so that no horizon can push transient information into it. Cells are SEP at each model's own best λ over $\{0, 1, 4, 16, 64\}$, $n = 5$ seeds; higher is better; differences within ± 0.005 SEP are called a tie. The asymmetric gate of Eq. 3 is better in 4 of 6 settings against 1 for the symmetric one, so the design choice is not an oversight.

Dataset	Stem	Asymmetric (ours)	Symmetric	Difference	Better
Synthetic	HGLP-Reg	0.814 ($\lambda = 1$)	0.673 ($\lambda = 1$)	-0.141	asym
Synthetic	HGLP-NCE	0.849 ($\lambda = 16$)	0.873 ($\lambda = 4$)	$+0.024$	**sym**
PTB-XL	HGLP-Reg	0.495 ($\lambda = 4$)	0.438 ($\lambda = 1$)	-0.057	asym
PTB-XL	HGLP-NCE	0.567 ($\lambda = 4$)	0.567 ($\lambda = 16$)	-0.000	tie
HAPT	HGLP-Reg	0.677 ($\lambda = 1$)	0.575 ($\lambda = 4$)	-0.102	asym
HAPT	HGLP-NCE	0.626 ($\lambda = 64$)	0.598 ($\lambda = 64$)	-0.028	asym

A Design choices, tested

Four choices in Section 4.1 could each have been made the other way. Each is ablated with everything else held fixed.

A.1 The gate is smooth, and one-sided

A step gate, $\mathbf{1}[\Delta < \tau]$, zeroes z_{mix}'s gradient outright beyond the threshold rather than passing a partial one. It is worse on both stems (Table 6). Since τ cannot be chosen accurately without labels (Section 7), horizons near it should not flip between fully passed and fully blocked under the smallest movement of the threshold; the width κ is what gives them intermediate values, and it earns its place.

The symmetric variant multiplies z_{per} by $1 - g(\Delta)$, switching it off below τ so that no horizon can deposit transient information in it. This is the obvious structural way to buy the exclusion the objective cannot enforce, and it does not work: it is better in 1 of six settings and worse in 4 (Table 7), costing 0.141 SEP on the synthetic regression stem. Removing z_{per} from the near horizons removes its training signal there as well, and the block loses more from being untrained than it gains from being protected — which is the argument of Section 3.2, now measured.

A.2 The target's receptive field is bounded

The gate rests on a premise: withholding z_{mix} at long range should cost nothing, because the transient factor is already forgotten there. That premise is a property of the data *and of how the target is defined*, so it can be measured directly. For each horizon we fit two linear readings of the target embedding, one from the persistent factor alone and one from both factors, and take the difference (Table 8).

With the receptive-field bound of Eq. 1 the difference reaches zero at $\Delta = 16$ and stays there. With the cumulative target of the earlier design — the causal encoder simply read at $t + \Delta$ — it never reaches zero at any horizon, because that target's receptive field runs back to the start of the window and therefore re-contains the anchor's own transient state. The shortcut is closed by the bound, not by hope.

Two limits on how far this may be read. It measures what explains the *target embedding*, not downstream error, and it reports only the *increment* the transient factor adds over the persistent one,

Table 8: Is withholding the transient block harmless at long range? Cells are $R^2(s, u \to \bar{z}_{t+\Delta}) - R^2(s \to \bar{z}_{t+\Delta})$: how much the transient factor adds, *beyond* what the persistent factor already explains, to a linear reading of the target embedding Δ ahead. **Zero means the transient factor is redundant there**, so gating it away costs nothing. Synthetic, HGLP-Reg, 2,500 steps, one seed, ground-truth factors. The bounded target reaches zero at $\Delta = 16$ — the value of τ used on this data — and stays there; the cumulative target never does, because its receptive field runs back to the start of the window and so re-contains the anchor's own transient state. This is the measurement behind the receptive-field bound of Eq. 1. It reads the target embedding rather than downstream error, and it is one seed on one dataset.

Target Δ	8	12	16	24	32	48	64	96	128
Bounded (ours)	+0.103	+0.025	+0.003	-0.001	-0.000	+0.000	-0.001	-0.000	-0.000
Cumulative	+0.024	+0.022	+0.024	+0.020	+0.017	+0.013	+0.011	+0.008	+0.008

On the same two runs, the bounded target also leaves less of the transient factor in z_{per} (R^2 0.041 against 0.660) and does not partially collapse (target RankMe 12.9 against 5.3).

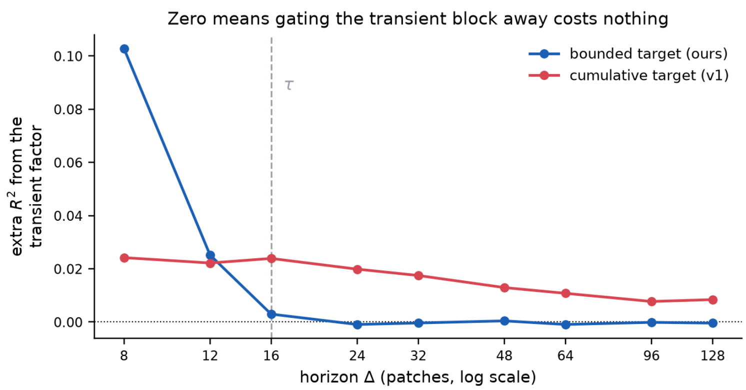

Figure 6: The same measurement as Table 8, as a curve. The bounded target falls to zero by $\Delta = \tau$ and stays there, so beyond the threshold the transient factor explains nothing about the target that the persistent factor does not already explain — which is what makes gating it away harmless. The cumulative target of the earlier design stays above zero at every horizon, including the longest: its receptive field reaches back to the start of the window and so re-contains the anchor's own transient state, and no horizon is far enough to escape that.

Table 9: Latent target against raw target. Cells are SEP on synthetic, $n = 5$ seeds, each stem at its own best λ. Bold marks the latent target. Everything but the prediction target is held fixed.

Stem	Latent (best)	Raw (best)	Difference
HGLP-Reg	**0.809** ($\lambda = 1$)	0.415 ($\lambda = 4$)	$+0.394$ ($1.95\times$)
HGLP-NCE	**0.844** ($\lambda = 16$)	0.432 ($\lambda = 16$)	$+0.411$ ($1.95\times$)

so it does not by itself establish that the persistent factor explains the target well. And it is one seed on one dataset, with ground-truth factors that only synthetic data provides.

A.3 The target is latent, not raw

This ablates the premise of the latent-prediction family: that prediction should be made in a latent space rather than on the raw signal. Gate, anchor, horizon and loss form are held fixed and only the prediction target changes. The latent target is the 64-dimensional output the target encoder reads from the future interval; the raw target is the patch itself at that time, with one extra linear head. Both models carry the cross-covariance penalty, so λ means the same for each, and each is compared at its own best value.

Predicting raw patches halves SEP on both stems (Table 9). The allocation term is the same for the two targets, at 0.86 to 0.89, and the gap opens in inclusion: the raw target loses the persistent

Table 10: Where the shortfall against SFA sits. Every cell is *ours minus SFA* on one term of SEP, paired by seed and then averaged, $n = 5$; positive means we lead. Our column is taken at the λ that maximizes our SEP, which is why λ is listed. Inclusion is a draw everywhere; **every real-data loss is a loss on the exclusion term alone** — the one term the objective does not enforce.

Dataset	Stem	λ	Inclusion	Allocation	Exclusion	SEP
Synthetic	HGLP-Reg	1	$+0.102$	$+0.236$	$+0.156$	$+0.395$
Synthetic	HGLP-NCE	16	-0.006	$+0.327$	$+0.266$	$+0.451$
PTB-XL	HGLP-Reg	4	-0.007	$+0.048$	$\mathbf{-0.136}$	-0.073
PTB-XL	HGLP-NCE	4	-0.000	$+0.007$	$\mathbf{-0.051}$	-0.031
HAPT	HGLP-Reg	1	-0.033	-0.011	-0.007	-0.039
HAPT	HGLP-NCE	64	-0.041	$+0.155$	-0.016	$+0.100$

Table 11: What survives a non-linear reader. The features, the splits and the blocks are held fixed and only the probe head changes, from a linear model to a one-hidden-layer MLP (256 units); the *same* head is given to SFA, PCA and ICA, so no method is tested more harshly than another. $\lambda = 4$, one seed. **(a)** Cells are MLP $-$ linear on one SEP term, averaged over four methods $\times$ six settings. Only exclusion falls, and it is the term the objective never enforced. **(b)** Cells are SFA's exclusion minus ours, so positive means SFA holds the lead; the diagnosis of Table 10 was not an artifact of linear probing.

(a) Change in each SEP term when the probe head stops being linear

SEP term	Mean change	Worst	Best	Rows
Inclusion	$+0.015$	-0.027	$+0.113$	16
Allocation	$+0.087$	$+0.031$	$+0.156$	16
Exclusion	-0.149	-0.530	$+0.019$	16

(b) Exclusion gap, SFA minus ours

Setting	Linear probe	MLP probe
PTB-XL/HGLP-Reg	$+0.085$	$+0.122$
PTB-XL/HGLP-NCE	$+0.020$	$+0.033$
HAPT/HGLP-Reg	$+0.009$	$+0.236$
HAPT/HGLP-NCE	$+0.000$	$+0.000$

factor as the penalty grows, falling from 0.807 to 0.401, whereas the latent target stays above 0.97 throughout. The raw signal still contains the transient component, and the pressure to predict that too pushes the persistent factor out. The gate and the penalty are thus not sufficient on their own; the latent target has to be present as well.

B Term decomposition and the non-linear probe

Table 10 splits SEP into its three factors and pairs by seed. Inclusion is a draw everywhere; every real-data loss against SFA is a loss on exclusion alone, and every win is a win on allocation. This matches the design rather than surprising it — the objective enforces availability and cannot enforce absence (Section 4.2), and it is on that term, and only that term, that the method falls behind.

Table 11 replaces the linear reading head with a one-hidden-layer MLP, giving the same head to every method. Part (a) reports the change in each SEP term: exactly one collapses, and it is exclusion. Part (b) reports SFA's exclusion lead under both heads: it survives, and in two settings widens, so the diagnosis of Table 10 was not an artifact of linear probing. PCA and ICA degrade sharply under the same head while SFA does not, which is worth noting on its own — the baseline we trail is a robust one.

Three cautions. Because exclusion falls monotonically as the reading head grows, only comparisons made under a fixed head are meaningful; the absolute value is a property of the probe as much as of the embedding. We tried one hidden width, 256, and did not trace the curve. And this experiment fixes $\lambda = 4$ with one seed, whereas Table 10 takes each setting at its own best λ, so cells of the two tables must not be read against each other.

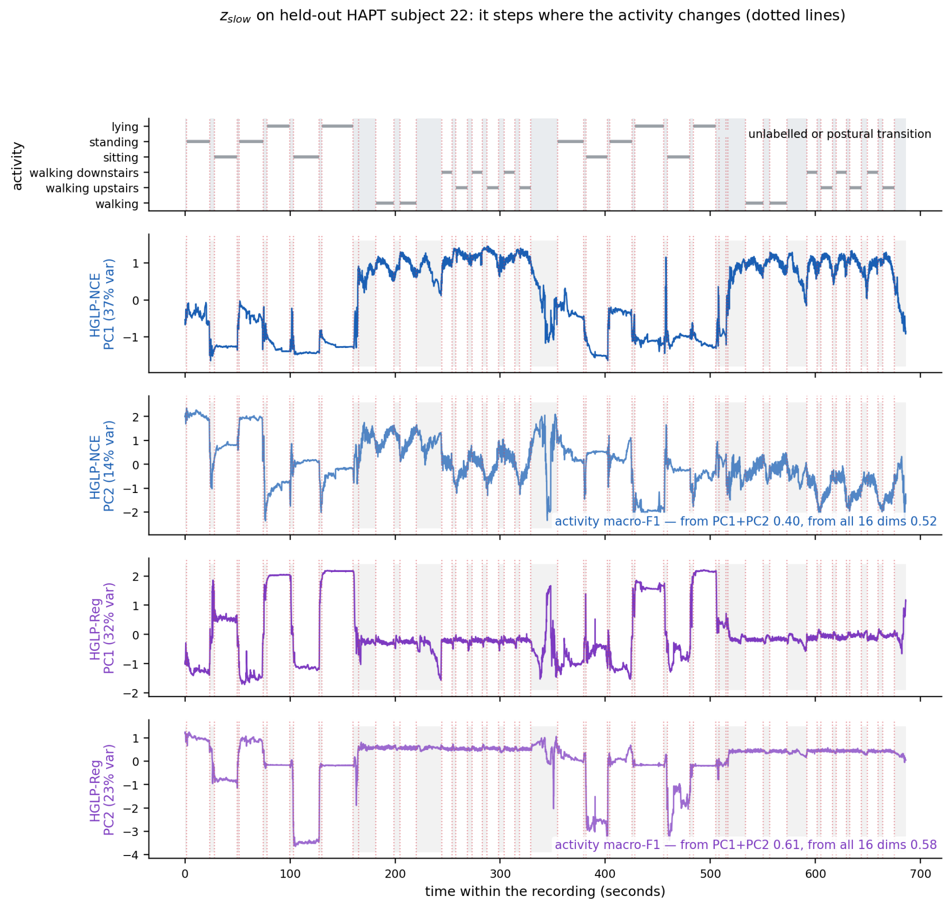

Figure 7: z_{per} read as a trajectory rather than as classifier input (HAPT). Using the separated block this way needs no labels at evaluation time and no probe, which is the use the downstream cost of Section 5.5 is paid for; we report the figure and do not claim a quantitative result from it.

C READING z_{per} AS A TRAJECTORY